

Deep Neural Network Based Multipath Mitigation Method for Carrier Based Differential GNSS Systems

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ABSTRACT

Carrier based Differential Global Navigation Satellite System (CD-GNSS) is getting lots of attention as a promising technology for drones since it can provide centimeter-level accuracy. Unlike widely used CD-GNSS applications such as geodetic survey, CD-GNSS systems to be used for drone applications should provide a certain level of integrity along with accuracy. However, there is a critical challenge in guaranteeing the integrity of CD-GNSS systems for drone applications due to a long filter duration which is the time necessary to resolve cycle ambiguity correctly. One of the most dominant factors that limits reducing the filter duration is the code multipath error. In this response, this paper proposes a code multipath mitigation method using Deep Neural Network (DNN) for drone systems. It is well known that DNN is a powerful for nonlinear regression problems toward which multipath estimation is applicable and can be trained using a large quantity of data. The target scenario of this study is chosen as a drone flying at a sufficiently high altitude (where no multipath reflections exist from surrounding obstacles) before lowering its altitude for other phases of operation. Therefore, the dominant factor of multipath errors of the drone is the signal reflection from its body frame. Considering the fact, the input parameters were chosen for developing the DNN model as elevation, azimuth and tilt angle of antenna which characterize signals reflected by the frame and Signal to Noise Ratio (SVR) which characterizes the slight change of signals. The validity of the proposed model was investigated and its multipath mitigation performance was evaluated under the target scenario using real data. The results show that the multipath error is mitigated by about 30% in terms of the standard deviation of multipath error and the filter duration is reduced by about 66%.

1.0 INTRODUCTION

As the drone industry and technology grow rapidly, many applications of drones are being proposed including tower inspection operating within Visual Line Of Sight (VLOS) or deliveries operating Beyond VLOS (BVLOS) [1]. The Unmanned aircraft system Traffic Management (UTM) system for both VLOS and BVLOS flights is in the development stage to enable efficient and safe drone operation in the low altitude (under 400ft) uncontrolled airspace. In order to operate many Unmanned Aircraft Vehicles (UAVs) in a limited space or to support new applications such as autonomous UAV refueling systems or landing systems, one of the most essential technologies required is a navigation system with a high level of accuracy. Real Time Kinematic (RTK), one of Carrier based Differential Global Navigation Satellite Systems (CD-GNSS), is gaining attention as a promising technology for drones since it can provide centimeter-level accuracy. However, CD-GNSS systems to be used for drone applications should provide a certain level of integrity along with accuracy, whereas RTK widely used for geodetic survey was originally developed for high accuracy applications without considering integrity [2].

There are challenges in guaranteeing the integrity of CD-GNSS systems. The precise navigation estimates obtained using differential carrier phase measurements depend on the successful resolution of cycle ambiguities and the integrity of the cycle resolution process should be guaranteed for safety critical systems [3]. The integrity of cycle resolution is a function of the quality of the measurements, satellite geometry, and filter duration (i.e., the time necessary to resolve integer ambiguity correctly). It is well understood that longer filter duration with better quality of the measurements yields better performance of cycle resolution [4]. In other words, with the low quality measurements, a much longer filter duration is required to achieve the same level of cycle resolution integrity. For example, a commercial RTK system if used for drone applications require about 67 minutes of filter duration to guarantee a Probability of Hazardous Misleading Information (P_{HMI}) of 10^{-7} for a given Vertical Alert Limit (VAL) of 1.1m and the

standard deviation of raw code multipath error of 30cm. This longer filter duration when compared to that of the Shipboard-Relative Global Positioning System (SRGPS) [3] that has the same integrity requirements is due to large uncorrected ranging errors resulting from the low cost receiver and antenna used in both the RTK reference stations and drone. Since such a long filter duration could be a limiting factor for drone operations, it is necessary to decrease the filter duration by improving the quality of measurements, especially by mitigating code multipath errors which is one of the dominant error sources of CD-GNSS systems.

The multipath errors are caused by time delay on Global Navigation Satellite System (GNSS) measurements induced by reflected signals. The signals reflected from ground or other objects often result in one or more secondary paths which are combined with the direct signal [5]. Since multipath errors vary depending on receiver/antenna, hardware platform where receiver/antenna are installed, and multipath environment, the correlation of multipath between two receivers operating at two different locations is often low. Thus, they are not removed by a differential technique, unlike ionospheric delay, tropospheric delay, and satellite orbit/clock errors which are highly correlated if two receivers in close proximity are used. Multipath errors account for a large portion of remaining errors in the carrier-based differential systems. Many studies have been conducted to reduce multipath errors over the past three decades [5]. The several existing multipath mitigation methods use benefit from signal propagation geometry such as the use of special antennas (e.g. the choke ring antenna) or spatial processing with multi-antenna arrays. The other methods take advantages of real time signal processing within receiver, such as using narrow correlator or different types of correlator functions [5]. However, many of those methods are undesirable because the antenna and the receiver are expensive or heavy to install on low cost drones.

Multipath error estimation methods using additional information (e.g., elevation angles or signal to noise ratio) which can be obtained from receivers have been studied as approaches without additional antennas or receiver processing [6] [7]. The method in [6] take advantage of the fact that the multipath error repeats day by day when the surrounding environment and user location do not change. The data of the previous day was used to estimate the multipath error using an adaptive filter in real time. For the drone which is in flight phase at a certain altitude (i.e., above ground objects which cause multipath), it can be assumed that there is no change in the surrounding environment. However, yesterday's time series of multipath error do not repeat today when the drone moves along a different path. Thus, the adaptive filter based method which is fully dependent on the time series of yesterday's multipath is not useful for drone applications. The pre-trained Support Vector Regression (SVR) using elevations and azimuths as inputs was studied in [7] to estimate and compensate for multipath errors. This method can be applied to not only stationary receivers but moving platforms, but it has a limitation to training a large number of data sets. The multipath of drone in motion can change as the orientation of the body frame of the drone with respect to satellites varies. This will increase the size of data sets and the complexity of SVR.

Therefore, this study proposes a Deep Neural Network (DNN)-based multipath mitigation method. It is possible to train a large quantity of data by adjusting the number of nodes of a Neural Network (NN) and also, it is well known that an NN is powerful for nonlinear regression problems toward which multipath estimation problem is applicable. Since the purpose of our research is to estimate the multipath errors included in raw measurements in real time, the output of the proposed model is a multipath error and inputs which well represent the characteristics of multipath errors should be available in real time. The dominant factor of multipath errors of drones in flight phase at high altitude is the reflected signals from its body frame. The input parameters were therefore chosen as satellite elevation and azimuth angles and tilt angle of antenna which characterize GNSS signals reflected by the frame. Signal to noise ratio (SNR) was also used as an input to characterize the slight changes of signals. The elevation, azimuth and SNR are available from a GNSS receiver and the tilt angle of antenna is obtained from IMU sensors in real time. To evaluate the performance of the proposed model, two case studies are conducted; 1) to investigate the validity of the DNN-based multipath mitigation method, and 2) to analyze the performance of the method under the virtual scenario for drone operations.

In Section 2, the motivation of this study is presented through the analysis of filter duration with respect to the integrity requirements and measurement quality. Section 3 introduces the DNN-based multipath mitigation method and describes the reasons why each input parameters is selected. Also, the validity of proposed model is investigated in Section 3. Section 4 provides the results of the performance evaluation of the proposed model, and finally, conclusions are given in Section 5.

2.0 ANALYSIS OF FILTER DURATION

In this section, the filter duration assuring cycle resolution integrity is analyzed with respect to the integrity requirements and quality of measurements for CD-GNSS systems. First, the observation model of CD-GNSS systems is introduced briefly and then

the relation between filter duration, integrity requirements, and measurement quality is discussed. Finally, the results of a simple analysis of filter duration with given integrity requirements and measurements quality are presented.

2.1 Observation Model of CD-GNSS Systems

We consider CD-GNSS which consists of the ground station and users and uses double difference measurements. The raw carrier phase measurements of GNSS are composed of true geometric range between a satellite and a receiver, cycle ambiguity, and other error sources which are the satellite and receiver clock bias, ionospheric and tropospheric delay and multipath errors [8]. The single difference measurement is defined as the difference in the raw measurements between the ground station and user of a common satellite. Further, the difference between the two single difference measurements of two satellites is referred to as the double difference measurement. The double difference measurement, ϕ_{ur}^{ij} , includes the double differenced range (r_{ur}^{ij}), cycle ambiguity (N_{ur}^{ij}), residual errors (I_{ur}^{ij} and T_{ur}^{ij}) resulting from spatial decorrelation, and multipath and noise (v_{ur}^{ij}) [8]. The same definitions hold for double difference code measurements, ρ_{ur}^{ij} , except no cycle ambiguity and the opposite sign of ionospheric delay as shown in Equation (1).

$$\begin{aligned}\lambda\phi_{ur}^{ij} &= r_{ur}^{ij} - I_{ur}^{ij} + T_{ur}^{ij} + \lambda N_{ur}^{ij} + v_{\phi,ur}^{ij} \\ \rho_{ur}^{ij} &= r_{ur}^{ij} + I_{ur}^{ij} + T_{ur}^{ij} + v_{\rho,ur}^{ij}\end{aligned}\quad (1)$$

The satellite and receiver clock bias are eliminated by differential operation. λ is the wave length, and the subscripts, u and r , represent the user and the ground station, respectively, and superscripts, i and j , represent two satellites each. The double difference measurements have the advantages of reducing the number of estimates and removing the correlated errors.

In turn, an additional difference between L1 and L2 measurements generates a new measurement which has a longer wave length than L1 and L2 [8]. The longer wavelength is useful for cycle resolution. The new measurement is known as the wide lane measurement. The final expression of the measurements, Z_w , to estimate double difference wide lane cycle ambiguity, N_w , is shown in Equation (2) and more details are in [4].

$$\begin{aligned}Z_w &= N_w + v_w \\ Z_w &= (\phi_{ur}^{ij}{}_{L1} - \phi_{ur}^{ij}{}_{L2}) - \left(\frac{\lambda_{L1} - \lambda_{L2}}{\lambda_{L1} + \lambda_{L2}} \right) \left(\frac{\rho_{ur}^{ij}{}_{L1}}{\lambda_{L1}} - \frac{\rho_{ur}^{ij}{}_{L2}}{\lambda_{L2}} \right) \\ N_w &= (N_{ur}^{ij}{}_{L1} - N_{ur}^{ij}{}_{L2}) \\ v_w &= \left(\frac{v_{\phi,ur}^{ij}{}_{L1}}{\lambda_{L1}} - \frac{v_{\phi,ur}^{ij}{}_{L2}}{\lambda_{L2}} \right) - \left(\frac{\lambda_{L1} - \lambda_{L2}}{\lambda_{L1} + \lambda_{L2}} \right) \left(\frac{v_{\rho,ur}^{ij}{}_{L1}}{\lambda_{L1}} + \frac{v_{\rho,ur}^{ij}{}_{L2}}{\lambda_{L2}} \right) \sim N(0, \sigma_w^2)\end{aligned}\quad (2)$$

where v_w is the remaining measurement error which induces the estimation error of N_w . Under the assumption that the variances of the carrier multipath and noise of L1 and L2 are equal and the same assumption holds for code multipath and noise, the variance of v_w is expressed as follows.

$$\sigma_w^2 = \left(\frac{1}{\lambda_{L1}^2} + \frac{1}{\lambda_{L2}^2} \right) \left(\sigma_{DD,\phi}^2 + \left(\frac{\lambda_{L2} - \lambda_{L1}}{\lambda_{L2} + \lambda_{L1}} \right)^2 \sigma_{DD,\rho}^2 \right) \quad (3)$$

where $\sigma_{DD,\phi}$ and $\sigma_{DD,\rho}$ represent the variance of the multipath and noise of double difference carrier and code measurements, respectively. The variance of raw code multipath and noise is about 30 cm, which is much larger than the raw carrier multipath and noise of 1cm. Therefore, the accuracy of estimation of N_w is greatly affected by the code multipath and noise.

2.2 Integrity of Cycle Resolution

The total integrity risk of CD-GNSS is the sum of the integrity risk under a Correct Fix (CF) of cycle ambiguities and that under an Incorrect Fix (IF) of cycle ambiguities as expressed in Equation (4). P_{IF} and P_{CF} are the probabilities of IF and CF, respectively [3]. The probability of Position Error (PE) exceeding its Protection Level (PL) should be less than the required Probability of Hazardously Misleading Information (P_{HMI}).

$$P(|PE| > PL | CF)(1 - P_{IF}) + P(|PE| > PL | IF)P_{IF} < P_{HMI} \quad (4)$$

Under the conservative assumption that PE is always larger than PL given the IF condition, we obtain Equation (5) from Equation (4). Accordingly, the Vertical Protection Level (VPL) for the nominal case is computed as Equation (6), where $k(x) = Q^{-1}\left(1 - \frac{1}{2}x\right)$, and Q is the standard normal cumulative distribution function [3].

$$P(|PE| > PL | CF) < \frac{P_{HMI} - P_{IF}}{1 - P_{IF}} \quad (5)$$

$$VPL = k\left(\frac{P_{HMI} - P_{IF}}{1 - P_{IF}}\right)\sigma_{vertical\ position\ error|CF} \quad (6)$$

To guarantee the cycle resolution integrity, it is desired that the P_{IF} be smaller than a level of required probability. The threshold of P_{IF} is determined from Equation (6) when other integrity requirements, P_{HMI} and VPL, are given. Since P_{IF} is a function of filter duration, T , and measurement accuracy, σ_w described in Section 2.1, as shown in Equations (7) and (8) [9], the required filter duration can be determined if the integrity requirements and σ_w are given.

$$\sigma_{\hat{N}}^2 = \frac{1}{T} \int_{-T}^T \left[1 - \frac{|\tau|}{T}\right] \sigma_{w,k} e^{-\beta|\tau|} d\tau \quad (7)$$

$$P_{IF} = 1 - \prod_{i=1}^{n-1} \left[2Q\left(\frac{1}{2\sigma_{\hat{N}_{||}}}\right) - 1\right] \quad (8)$$

where $\sigma_{\hat{N}}^2$ is the variance of estimated integer ambiguities, k represents each epoch within filter duration T , N is the cycle ambiguity and β is the time constant of multipath error modeled as a first-order Gauss-Markov random process [9]. In this paper, β is assumed as 1 minute [9]. Since σ_w is greatly affected by $\sigma_{DD,\rho}$ (as explained in Section 2.1), the smaller $\sigma_{DD,\rho}$ yields the shorter T for a given P_{IF} . Therefore, if we mitigate the code multipath error, we can reduce filter duration while maintaining the required P_{IF} .

2.3 Filter Duration with Respect to P_{HMI} and Code Multipath

This subsection conducts a sensitivity analysis to analyze the quantitative aspects of filtering duration with respect to integrity requirements and quality of measurements. The filtering duration with respect to quality of measurements was analyzed for SRGPS in [3], but no analysis has been done with respect to integrity requirements. Since the regulations and system requirements concerning drone applications are still in the early stages of development, analysis of the relationship between filtering duration and integrity requirements would be beneficial to determine system requirements according to specific drone applications. Therefore, multi-dimensional analysis was carried out on filtering duration as a function of P_{HMI} and code multipath.

Several assumptions were applied to make the analysis simple. First, we used a fixed VAL of 1.1m which is the requirement of SRGPS. The variance of double difference code multipath error was used as measurement quality because it is the dominant source as explained in Section 2.1. To obtain a conservative result, we used the variance of code multipath error at an elevation angle of 10° regardless of the actual satellite elevation. In order to directly express P_{IF} as a function of the variance of double difference code multipath error, this analysis utilized the Integer Bootstrapping (IB) algorithm with LAMBDA method for cycle ambiguity resolution [9]. The results from multidimensional analysis of filter duration with respect to P_{HMI} and the variance of double difference code multipath are shown in Figure 1. Although this is a simplified analysis with many assumptions, the results are in a good agreement with those obtained from the simulation in [4]. The simulation uses the constellation of Global Positioning System (GPS) and takes into account the changes of satellite geometry and the resulting changes in variance of multipath errors.

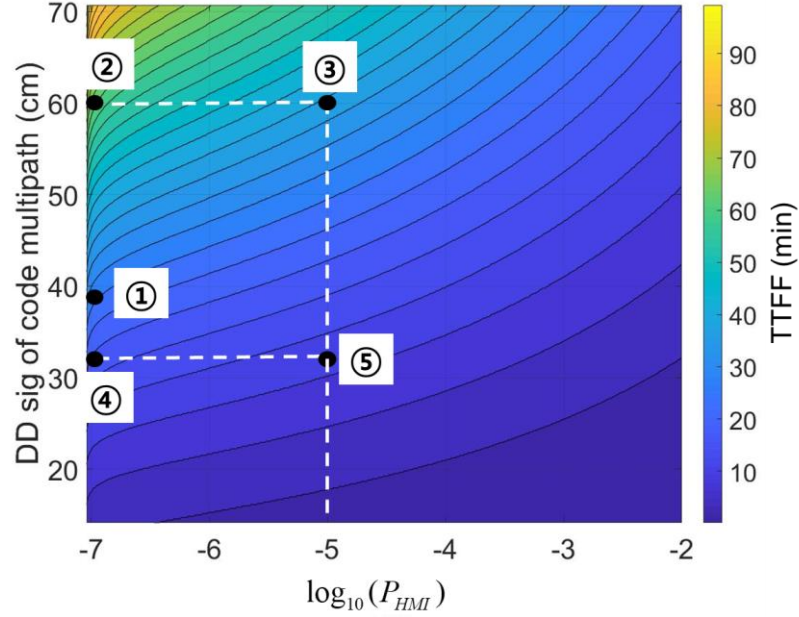


Figure 1. Results from multidimensional analysis of filter duration with respect to the required P_{HMI} and the variance of double difference code multipath error

The x- and y-axis represent the integrity requirement, P_{HMI} , and the variance of double difference code multipath error, respectively. The corresponding color represents the filter duration. Since the SRGPS (① in Figure 1) uses a state of the art antenna and receiver that can achieve about 38cm of the variance of double difference code multipath error and aims to guarantee a P_{HMI} of 10^{-7} , about 28 minutes of filter duration are required. This result is consistent with that in [4]. If a commercial RTK system (② in Figure 1) is used for a drone equipped with low cost antenna and receiver whose variance of double difference code multipath error is about 60cm, this system requires a filter duration of about 67 minutes to assure a P_{HMI} of 10^{-7} . Even if we relax the P_{HMI} to 10^{-5} (③ in Figure 1), it takes about 39 minutes. A navigation system that needs such a long filter duration is difficult to use in practice. A method of reducing the filter duration with a given level of integrity balanced with efficiency of drone operations is to improve the quality of measurements by mitigating code multipath errors.

3.0 NEURAL NETWORK BASED MULTIPATH MITIGATION METHOD

In Section 3, we propose a multipath estimation model to mitigate code multipath error using DNN. First, the overview of the proposed model is introduced. Second, input parameters are explained. Then, we explain how to extract multipath learning data from receiver observations and finally verify the validity of the proposed model.

3.1 DNN based Multipath Estimation Model

There are many previous studies about multipath mitigation methods for stationary receivers [5], [6], [7], but not for moving platforms such as drones. The purpose of this study is to estimate and compensate the code multipath error of moving drones in real time using a pre-trained model in order to improve the quality of code measurements. Thus, we propose a multipath estimation model based on DNN which is composed of input layer, hidden layers, and output layer as shown in Figure 2. The loss function of the proposed model is mean squared error, and the activation function is ‘tanh’, and the Levenberg-Marquardt algorithm is used to optimize the weights. We set up one hidden layer and 20 hidden nodes to validate the DNN model, and for scenario studies in Section 4, we used one hidden layer and 30 hidden nodes. The reasons for using smaller number of hidden nodes for validation are to show that each input is chosen well by reducing the complexity of the model and to make time required for training shorter. The numerical values of hidden layers and nodes in this paper are chosen as an example; in practice, they should be optimized according to the circumstances (e.g. body frame of drones, hardware configuration) through trial and error.

In order to design the model for estimating the multipath, it is important to determine the input parameters. As reflected signals cause multipath errors, the inputs should be informative parameters about signal reflection to characterize multipath errors well and should be available in real time. In this paper, we choose elevation and azimuth angle of satellite, Signal to Noise Ratio (SNR), and the antenna angle as inputs and the details are described in the following section.

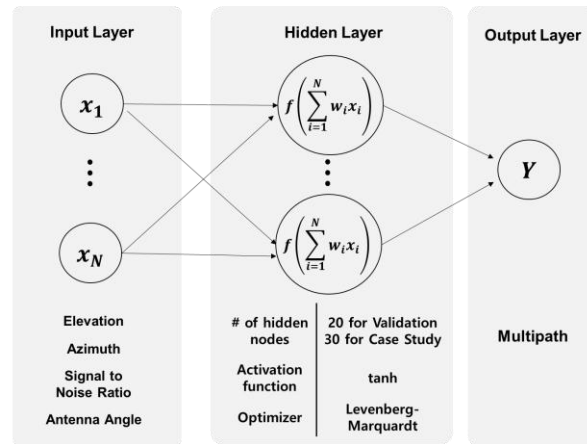


Figure 2. The structure of a Deep Neural Network model

3.2 Input Parameters

3.2.1 Elevation and Azimuth of Satellite

Multipath errors caused by signal reflections combined with a direct path signal are strongly related to receiver-satellite geometry which can be expressed as the elevation and azimuth angles of satellites. The elevation of the satellite is its angular height measured from the horizontal plane of the receiver and the azimuth is the angle measured from the north direction to the satellite on the horizontal plane. It is well known that the lower the elevation angle, the greater the multipath errors. That is because the signal transmitted by the low elevation satellite is easily blocked or reflected. In a previous study [7], the elevation and azimuth are used as to SVR for constructing a multipath regression model.

When drones are in cruising phase above ground objects, most multipath errors will be caused by the signals reflected by their body frame or equipment. Many pieces of equipment, including antenna, flight controller, and camera, are installed on the drone, and thus various secondary paths of the signal may occur depending on the direction from which the signal is received and the direction toward where the heading of the drone points. However, in this paper, we assumed that the hardware configuration of the drone is symmetric with respect to the central axis of the drone for simplicity. Therefore, the signal reflection depends on the elevation and azimuth angles only, and they are informative parameters to estimate multipath error. They are available in real time from the ephemeris data, and thus we selected them as inputs of the multipath estimation model.

3.2.2 Signal to Noise Ratio

In addition to the elevations and azimuth angles of satellite, we added the SNR, a sensitive parameter that can characterize the surrounding environments, as an input parameter. In [10], it is shown that both the multipath error and SNR exhibit a periodic behavior with respect to the difference of the distance, ζ , between the direct path and secondary path and they are in-phase. The relation between multipath error and SNR is derived analytically in [10] and expressed as

$$\begin{aligned} \text{SNR}(t) &= \text{SNR}_0 + \overline{\text{SNR}_e} - \frac{\alpha}{\beta} \overline{\text{MP}} + \frac{\alpha}{\beta} \text{MP}(t) \\ &= C(t) + \gamma(t) \text{MP}(t) \end{aligned} \quad (9)$$

where SNR_0 is the SNR of the direct signal, SNR_e is the difference of SNR between the direct path signal and secondary path signal, MP is code multipath error, and $\bar{x}$ represents the mean value of x . α and β are the amplitudes of SNR and MP, respectively, when modeled as $\cos(\zeta)$. Since the values of $C(t)$ and $\gamma(t)$ vary depending on the surrounding conditions, it is very difficult to derive them analytically. However, Equation (9) shows that the SNR and the multipath error have a strong correlation. Therefore, SNR is an informative input parameter for capturing changes of multipath that cannot be predicted by the elevation and azimuth angles only. SNR is observable and commonly recorded by receivers in real time, which made it a good input candidate.

3.2.3 Antenna Angle

The drone has to be tilted to move horizontally, and thus the antenna attached on the drone is also tilted as depicted in Figure 3 where the red ring represents the antenna gain. As shown in the figure, if the antenna is tilted, the gain pattern changes and that results in the change of the signal strength and multipath error. Furthermore, the body frame tilt affects not only the antenna gain pattern but also the signal reflections caused by the body frame and other surrounding objects, and consequently the difference in multipath errors due to the body frame tilt will be much larger. Therefore, to consider these characteristics, we set the antenna angle to an input parameter of the model.

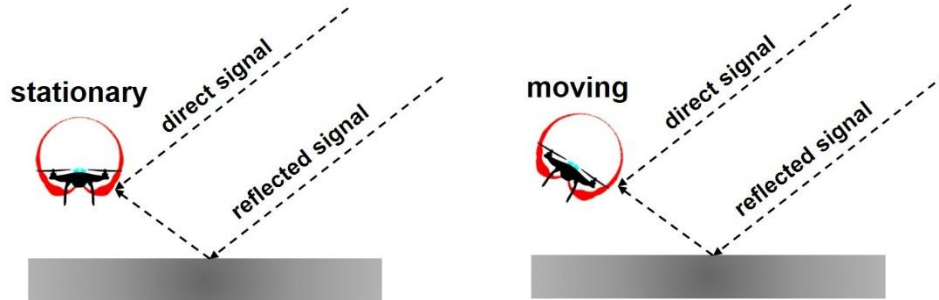


Figure 3. The antenna's gain pattern when the drone is hovering (left) and moving (right)

Figure 4 shows the multipath error of tilted and non-tilted antennas. The multipath for the non-tilted case is presented as the red curve and that of the 30° tilted antenna towards north direction is shown as the blue curve. Both multipath errors were estimated using datasets collected at the same position and the black curve shows the elevation angle of the satellite. When the satellite is at the elevation of between 15-30°, the difference between the non-tilted and tilted cases is clearly visible (Figure 4). Currently, most drones are required to be equipped with an Inertial Measurement Unit (IMU), and the attitude of drones, i.e., the tilt angle of the antenna, is available in real time using the IMU sensor. Thus, we can use the antenna angle as an input parameter in real time.

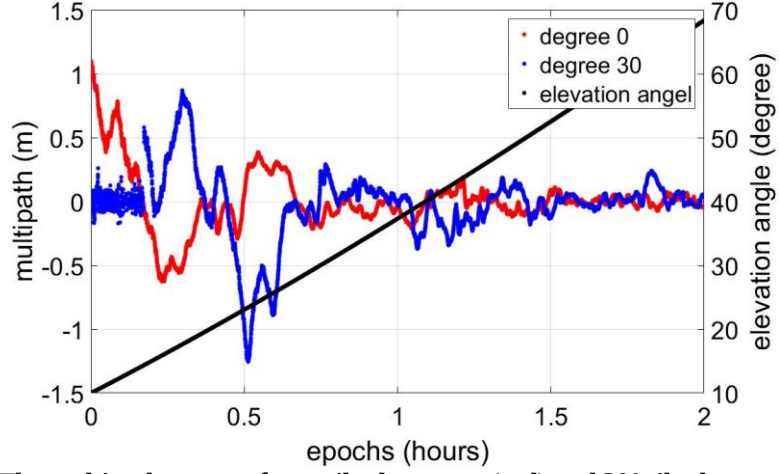


Figure 4. The multipath errors of non-tilted antenna (red) and 30° tilted antenna (blue)

3.3 Extracting Multipath Error

Multipath data are needed to train the model of multipath estimators that will be described in Subsection 3.4. The multipath error was extracted using Code-minus-Carrier (CmC) measurements which are generated by subtracting the raw carrier measurements from the raw code measurements as presented in Equation (10)

$$\begin{aligned}\chi &= \rho - \phi \\ &= 2I - N + v_\rho - v_\phi\end{aligned}\quad (10)$$

where χ is CmC, ρ and ϕ are the raw code and carrier measurements, I is the ionospheric delay, N is the cycle ambiguity, and v is the multipath error. The carrier multipath error, v_ϕ , is negligible because it is much smaller than the code multipath error, v_ρ . To isolate the code multipath error only, the ionospheric delay and cycle ambiguity term should be removed in Equation (10). The ionospheric delay can be directly computed using dual frequency measurements via the following equation where f is the frequency of the signal.

$$I = \frac{\rho_{L2} - \rho_{L1}}{f_{L1}^2 / f_{L2}^2 - 1} \quad (11)$$

As long as a cycle slip does not occur, the integer ambiguity can be regarded as a constant. Since the code multipath error is assumed to follow a zero mean Gaussian distribution, the cycle ambiguity can be obtained by averaging ionospheric free CmC measurements over a given orbit arc. The averaging should be reset if a cycle slip occurs. We can isolate the code multipath error by subtracting the cycle ambiguity from ionospheric free CmC at each epoch [7].

3.4 Validity of Proposed Model

To verify the validity of the proposed model, two case studies were conducted. One is for the effectiveness of input parameters and the other is for effectiveness of the model type.

3.4.1 Training and Test Data

We train a DNN-based multipath estimation model with real data and test the performance of the trained model. All of the data used in this study was collected through the antenna installed at the rooftop of the KAIST aerospace engineering department building

as illustrated in Figure 5. The Antenna A and B antennas are fixed and located on top of the 3 m pillar where no interference from other buildings is present, except signal reflections from the surface of the rooftop. The Antenna C antenna is attached on the handrail located at the edge of the building and the tilt angle can be adjusted. The altitude of the Antenna C antenna is lower than those of Antenna A and B antennas. All of the data were collected for two consecutive days; the data collected on the first day is used for training and data collected on the second day is used for testing. The input parameters (reviewed in the previous section) and multipath error obtained from the first day's data are used for the training. The multipath error is used as labeling data. The input parameters obtained from the second day's data and the model developed are used for multipath prediction, and the multipath errors extracted from the second day's data by using the method explained in Subsection 3.3 are used as true values for performance evaluation. The data collected from the Antenna A antenna is used for experiments to validate the proposed model, and the data collected from Antenna A, B and C is used for experiments that evaluate the performance of the model under target scenarios in Section 4.

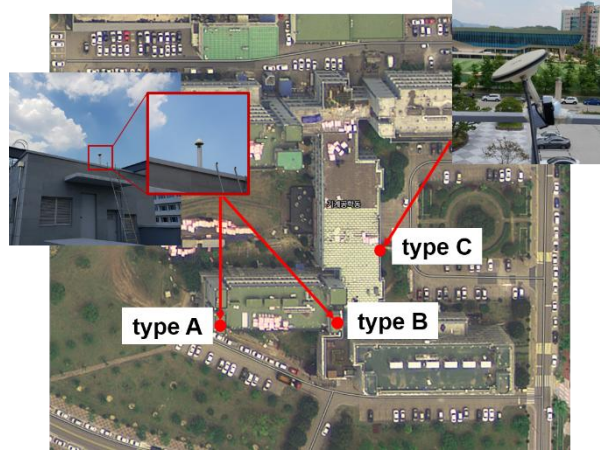


Figure 5. The hardware configuration used for performance evaluation

3.4.2 Effectiveness of Input Parameters

The inputs selected in Section 3.2 are satellite elevation, satellite azimuth, SNR, and antenna tilt angle. The elevation and azimuth angles are important parameters for modeling multipath errors, and their effectiveness was sufficiently verified through previous studies [7]. In addition, the difference in multipath errors according to antenna tilt angles has been confirmed in Subsection 3.2.3. Therefore, we analyze the effectiveness of the SNR only in this section. We evaluated the performance of two models, one that includes the SNR as input and another that does not.

As a result, when the SNR was not included in the inputs, the multipath error was reduced by about 14% in terms of variance and when the SNR was included, 32% of multipath was reduced. Figure 6 shows the results of multipath prediction. The left plot shows the case in which SNR is not included and the right plot shows the case including SNR. The blue curve is the true multipath error and the red curve is the predicted multipath error using the proposed model. As can be seen in Figure 6, when the SNR is not used, it converges to a specific value, while when using SNR, multipath error is estimated well over the whole interval. Therefore, SNR is important information and should be included as an input.

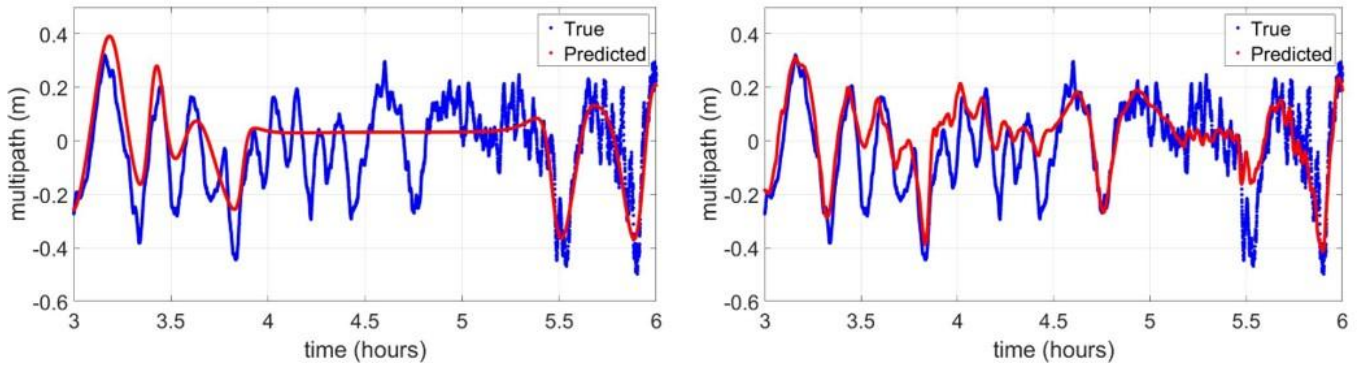


Figure 6. The results of multipath prediction using the proposed model without SNR (left) and with SNR (right)

3.4.3 Effectiveness with Respect to Model Type

The proposed model uses the basic DNN structure that is a kind of nonlinear regression model. There are a lot of nonlinear regression models that can be used to estimate the multipath error. Therefore, in this section, we evaluate the multipath estimation performance of various types of regression model, i.e., the SVM and the Recurrent Neural Network (RNN), and compared the results with the DNN-based model. We used three kinds of kernels, which are linear, second, and Gaussian kernels to train SVM. The RNN including 20 hidden nodes was constructed and trained, and also the DNN including the same number of hidden nodes was used.

The summary results of the performance evaluation are shown in Table 1. In the case of SVM, the performance is poor when using a linear kernel or a secondary kernel because multipath errors are highly nonlinear to be related as a simple linear or quadratic functions. Gaussian-kernel-based SVM reduced the multipath error by about 21%, but this is also less effective than the NN based model. The performance of the RNN-based model which processes the sequence of inputs is about 2% better than the DNN-based model. However, RNN takes an additional initial time which is the waiting time to get the first sequence of inputs and the training time is about 10 times longer than the DNN. Therefore, the DNN-based model, which does not need an initial time, has a fast training time, and suffers a small performance degradation compared with RNN, is the most appropriate to estimate multipath error.

Table 1. The summary of performance evaluation with respect to the model type

Model		Reduction Rate	Remarks
SVM	Linear kernel	1%	Lower performances than neural network model
	Quadratic kernel	3%	
	Gaussian kernel	21%	
RNN	Hidden nodes 20	35%	- Performance of RNN is the best - But RNN needs initial time and takes a long time to train model
DNN	Hidden nodes 20	33%	- Performance of DNN is little lower than RNN - But does not need initial time and fast training time

4 SCENARIO STUDY

In this section, the performance analysis is structured with example scenarios of precision navigation for drone flights. To validate the developed model performance using sufficient amount of data, we assumed flight scenarios which have a flight time of more

than 3 hours. Due to the limited battery capacity of small size drones for collecting more than 3 hours of consecutive flight data, data were collected from the ground that simulates the drone flight environment as similar as possible.

4.1 Target Scenario

Our target scenario is that a drone flying at a sufficiently high altitude (where almost no multipath reflections exist from surrounding obstacles) before landing for its missions as depicted in Figure 7. We also assumed that the drone resolves its cycle ambiguity before lowering its altitude to avoid performing the task under unexpected multipath (i.e., untrained multipath error) environments during the landing phase. Under the target scenario, two multipath conditions are considered; 1) Case 1: the low multipath environments where the signal is reflected mostly from the frame of the drone (not from buildings or other obstacles), 2) Case 2: the change of multipath error caused by change in drone attitude. The results of those cases are presented as follows.

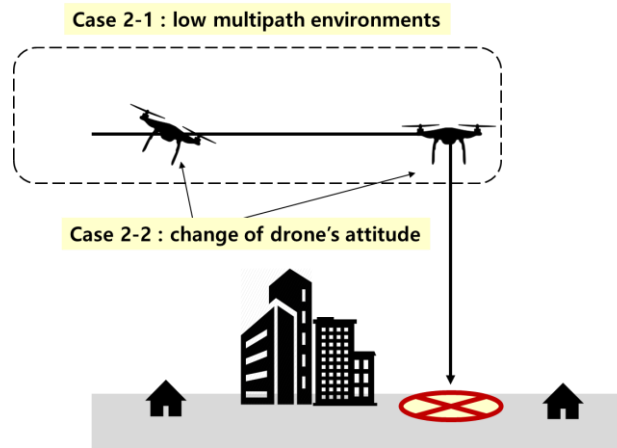


Figure 7. Target scenario for the performance tests

4.2 Case 1: Performance Test under Low-Multipath Environments

In this section, we analyze the multipath error estimation performance of the proposed model trained with the data under low multipath environments. The data were collected from Antenna A and B antennas. Both antennas are installed in low multipath environments where no multipath except those from the surface of rooftop exists) similar to that of drones flying at high altitudes, i.e., no multipath except those from the drone frame. Figure 8 shows the first day of multipath data collected using Antenna A (the black curve) and Antenna B (the blue curve) antennas, respectively.

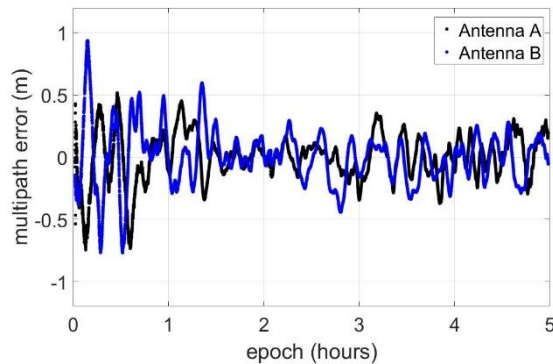


Figure 8. Multipath data collected using Antenna A (the blue curve) and Antenna B (the black curve) antennas

In order to train a model with the first day data collected from both Antenna A and B antennas, 30 hidden nodes were used and the ensemble method using 5 models was applied to prevent the overfitting problem and to improve its performance. The ensemble

method is the process of training multiple models using the same model structure and the same training data but different initial value of weights, and averaging the outputs of each model to produce the final output. We tested the performance of the trained model by applying it to predict the multipath errors at Antenna A and B antennas on the second day. The results are in Figure 9 where the red curves represent the estimated multipath errors using the trained model, and the black and blue curves are the true multipath errors for Antenna A and B antennas, respectively.

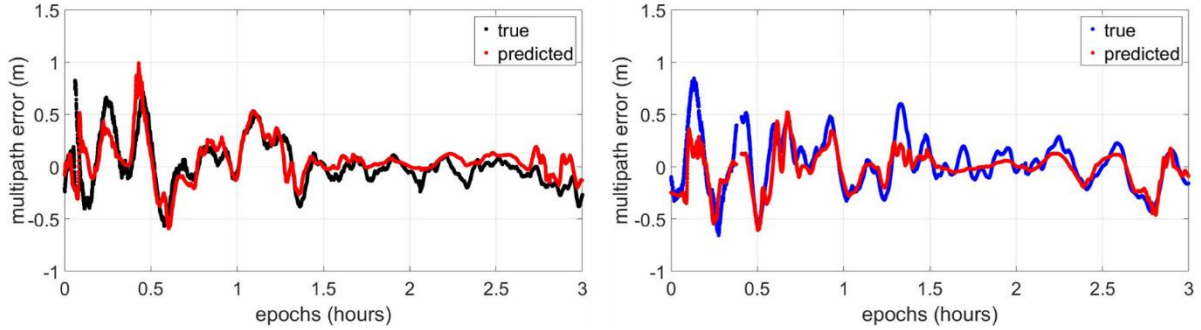


Figure 9. The results from multipath error estimation on Day 2 for Antenna A (left) and Antenna B (right) using the model trained with Day 1 data of both Antennas A and B

As shown in Figure 9, the multipath model trained with both Antennas A and B estimates the multipath errors of each antenna well by discriminating between Antenna A and Antenna B. The standard deviations of multipath error for Day 2 data before multipath error mitigation were 0.1929 m for Antenna A and 0.2067 m for Antenna B, respectively. If the multipath errors are calibrated using the trained model, the standard deviations of the residual error are reduced by about 30%, resulting in 0.1322 m for Antenna A and 0.1443 m for Antenna B.

The standard deviations of the non-mitigated and mitigated multipath errors with respect to the satellite elevations are depicted in Figure 10. The representation of the color is the same as Figure 9. As shown in Figure 10, the standard deviations decreased at most elevations, except those in the lowest elevation. The signal from the lowest elevation incurs lots of cycle slip, and consequently the extracted multipath data used for learning are not accurate because of short arc length. Accordingly, during the training process, the model could not find any correlation between the inputs and output, and as a result, the model was not trained properly at the lowest elevation. Thus, in practice, it is recommended that the cut-off angle of satellite elevation be larger than 10° .

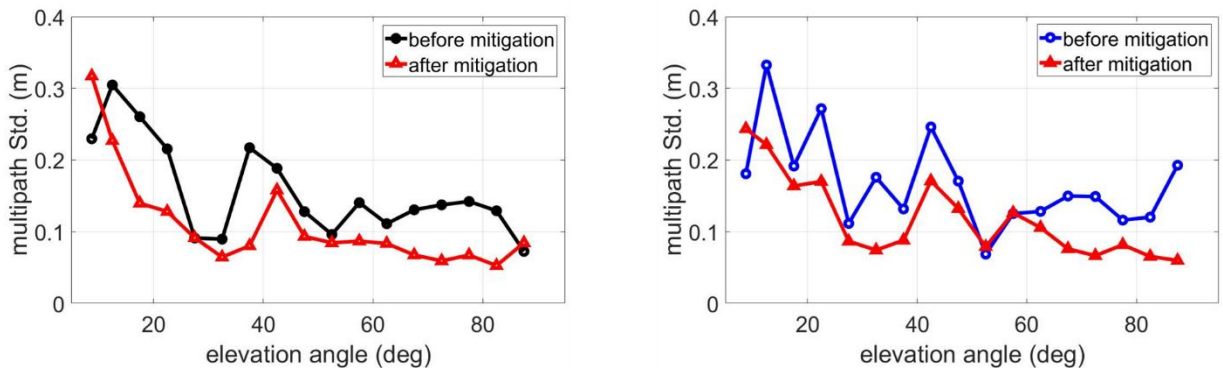


Figure 10. The standard deviations of multipath error for Antenna A (left) and Antenna B (right) antennas with respect to satellite elevation angle

4.3 Case 2: Performance Test for Drone Attitude Changes

The drone has to be tilted to move horizontally and the change of attitude induces the different multipath errors even with the same signal. In this section, we analyze whether the proposed model can estimate the difference multipath error caused by the attitude change of drones. Therefore, we trained a DNN-based model using data collected at Antenna C with different antenna tilt angles (i.e.,

0 and 30 degree tilt angles). Figure 11 shows the multipath error data collected from the antenna tilted by 0 (the blue curve) and 30 degree (the black curve), respectively.

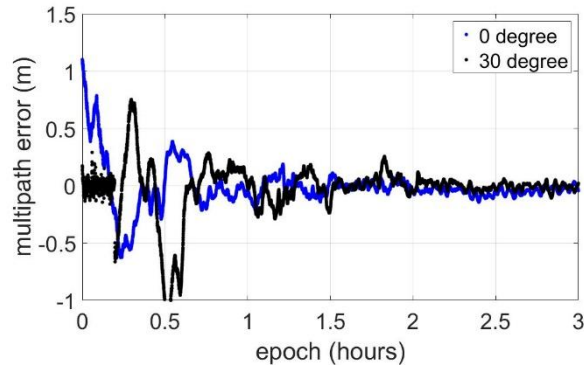


Figure 11. The multipath error data collected at the antenna tilted by 0 degree (the blue curve) and 30 degree (the black curve)

We trained the DNN model using 30 hidden nodes and Day 1 data, and tested the trained model using data collected on Day 2. The results of the test are depicted in Figure 12 where the blue and black curves represent the true multipath error for 0° and 30° tilt angles, respectively, and the red curves show the multipath errors estimated by using the trained model. The standard deviations of the multipath errors decreased by 42% (from 0.1665 m to 0.0951 m) for the case of 0° tilt angle, and 30% (from 0.1975 m to 0.1386 m) for the case of 30° tilt angle.

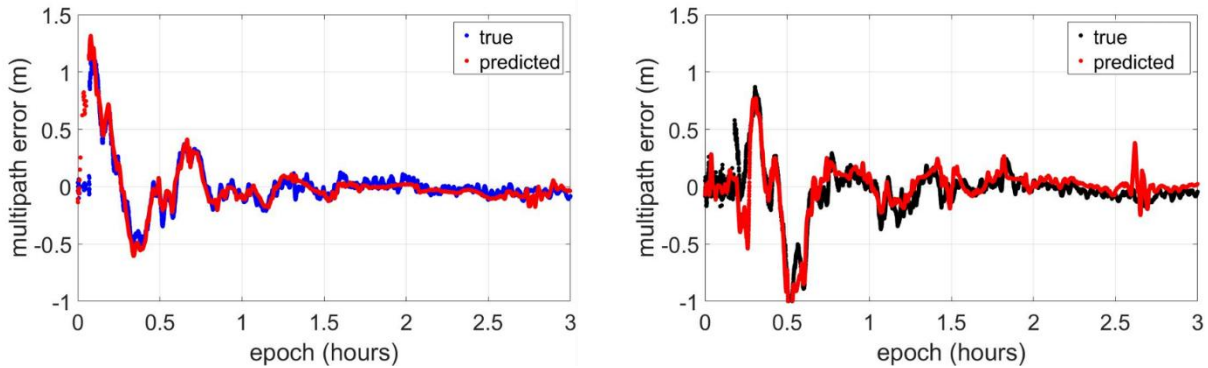


Figure 12. The results from multipath error estimation using the trained model when the antenna tilt angles are 0° (left), and 30° (right)

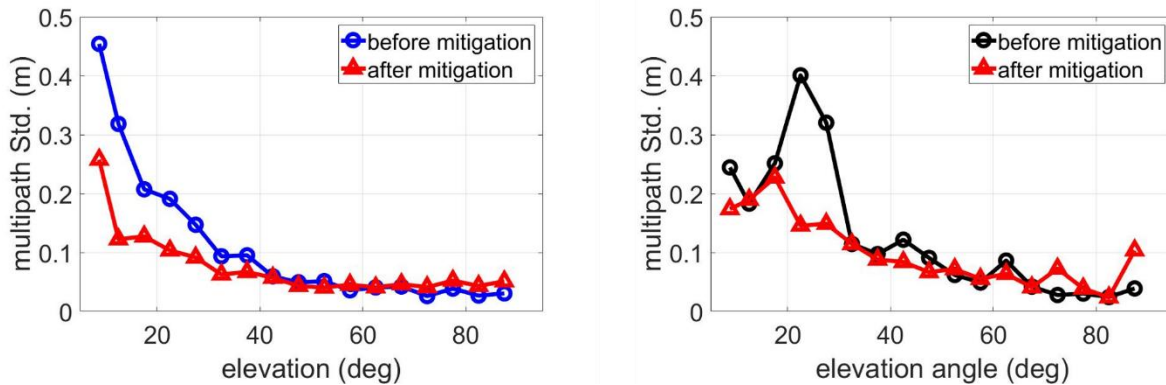


Figure 13. The standard deviation of multipath error with respect to satellite elevation angles under 0° (left) and 30° (right) antenna tilt angles

The standard deviations with respect to the elevation angle is depicted in Figure 13. The standard deviations were reduced at most elevation angles, but some increased at high elevation angles. That is because, in general multipath errors are mostly induced by the signal reflected from the ground, when the elevation angle of a satellite is high. However, since the Antenna C antenna is attached to the handrail located at the edge of the building, it is free from ground reflections. Therefore, in the case of the Antenna C antenna, there are fewer multipath but more random noise at high elevation. Since the random noises do not have any information about the relation between inputs and output, they interfere with training the model. Thus, the standard deviations increased at some high elevations.

Table 2 Summary of performance tests

Scenarios	Antennas used for training	Antennas used for performance evaluation	Performance Improvement (standard deviation of multipath error)
Case 1: (Performance Test for Low-Multipath Environments)	Antenna A & B	Antenna A	0.1929 m $\rightarrow$ 0.1322 m (31% reduction)
		Antenna B	0.2067 m $\rightarrow$ 0.1443 m (30% reduction)
Case 2: (Performance Test for Drone Attitude Changes)	Antenna C (0° and 30° antenna tilt angles)	Antenna C (0° tilt angle)	0.1665 m $\rightarrow$ 0.0951 m (42% reduction)
		Antenna C (30° tilt angle)	0.1975 m $\rightarrow$ 0.1386 m (30% reduction)

4.4 Use of Multipath Estimation Model to Reduce Filter Duration

In this section, we analyze the benefit of the proposed multipath mitigation method by comparing the filter durations before and after applying the mitigation method. The analysis were conducted under two conditions; 1) antenna mounted on top of a drone (reviewed through Section 4.3), 2) a reference station installed at a good multipath environment. Unlike the case of operating drones, a reference station is fixed at a location on the ground. Therefore, the multipath error has a periodic characteristic and is repeated every 24 hours as long as the surrounding environment stays constant. If yesterday's multipath data is available, then it is very powerful to use yesterday's data to mitigate multipath error today. In response, we applied yesterday's multipath data to mitigate multipath error at a reference station whereas the developed DNN-based model is applied for the drone case.

The standard deviations of raw code multipath error with respect to satellite elevation are shown in Figure 14. The data used for Figure 14 is collected using the Antenna A antenna. As explained in Section 2.3, we used the same standard deviation for any satellite elevation measurements as the value at 10°. Thus, the standard deviation reduced from 30 cm to 10 cm for the reference station and from 30 cm to 22 cm for the drone case. Finally, the double difference code measurement is reduced from 60 cm to 34 cm. Correspondingly, the filter duration is reduced from 66 minutes to 22 minutes when $P_{HMI} 10^{-7}$ is guaranteed (④ in Figure 1). Also, if the requirement of P_{HMI} is relaxed to 10^{-5} , the filter duration can be reduced from 39 minutes to 13 minutes (⑤ in Figure 1).

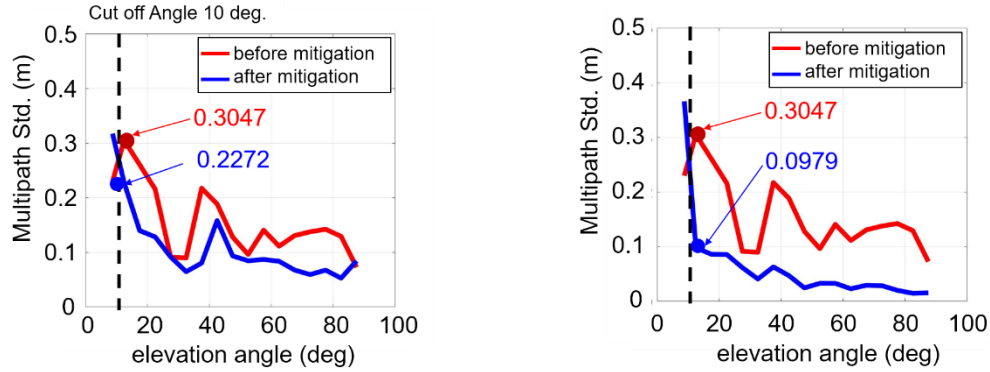


Figure 14. The standard deviation of raw code multipath error of a drone (left) and a reference station (right) with respect to the satellite elevations

5.0 CONCLUSION

We proposed a DNN-based multipath mitigation method in real time and elevation, azimuth, SNR, antenna angle were selected as inputs that can characterize the multipath errors well. The elevations and azimuths give the information of the satellite-receiver geometry. The SNRs characterize the slight changes of multipath errors that cannot be caught by elevations and azimuths. The antenna angles give information about the difference of multipath error induced by the change of attitude of the drones.

Various experiments were conducted to evaluate the performance of the proposed multipath estimation model. The data used for the analysis was not real flight data, but was collected by organizing as similar an environment as possible on the ground. First, to verify the validity of the proposed model, the effect of inputs and the effect of model type were analyzed. Second, the performance evaluation was conducted for two cases that drones can encounter. One situation covered low multipath environments and the other addressed the change in the drone's attitude. As a result, multipath errors are reduced about 30% in terms of the standard deviations. Also, the TTFF (time to first fix) was reduced by about 66% when the multipath errors were calibrated using the proposed model.

In this research, the configurations (i.e., the number of hidden nodes and layers, activation function, learning rates, batch size), of DNN used for performance evaluation were not optimal. Therefore, we expect the performance to improve if the configurations of DNN are optimized. Also, this research just proposes a possibility that the DNN-based model can be used to estimate the multipath error using the data collected on the ground. For future work, real flight experiments should be conducted.

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